...

Triple Integrals

Suppose a large solid box occupies the region $[0,2]\times[0,2]\times[0,1]$ in xyz-space, and that the density of the object is variable, modeled by $\rho(x,y,z)=\frac{10x+10y}{3-2z}$ in kg per cubic meter at (x,y,z) (assume x,y,z are in meters). Suppose we would like to estimate the mass of the object!

We can divide $[0,2]\times[0,2]\times[0,1]$ into 4 sub-cubes of volume $\Delta V=1$ cubic meter by slicing the object with the planes $x=1$ and $y=1$. Then we can use, as "sample points," sub-cube centers at $(0.5,0.5,0.5)$, $(1.5,0.5,0.5)$, $(0.5,1.5,0.5)$ and $(1.5,1.5,0.5)$ to estimate the mass:

$$m\approx\left(\frac{5+5}{3-2(0.5)}+\frac{15+5}{3-2(0.5)}+\frac{5+15}{3-2(0.5)}+\frac{15+15}{3-2(0.5)}\right)\cdot 1=40.$$

This is an example of a Riemann sum of a function of three variables.

Let $f(x,y,z)$ be defined over a solid region E in xyz-space. If the limit,

$$\lim_{n\to\infty}\sum_{i=1}^{n}f(x_i,y_i,z_i)\Delta V_i,$$

exists for all partitions of E into n sub-solids,where (x_i,y_i,z_i) is a sample point from the ith sub-solid, and ΔV_i (such that $\Delta V_i\to 0$) is the volume of the ith sub-solid, then the limit is the triple integral of f over E.

Notation:

$$\int\int\int_E f(x,y,z)dV.$$

Fubini's theorem: We can iterate the integral when f is continuous over E.

Let's see how good our mass estimate was: The mass of the large box is

$$\begin{aligned} m = 10\int_0^1\int_0^2\int_0^2 \frac{x+y}{3-2z}\,dx\,dy\,dz &= 10\int_0^1\int_0^2 \frac{0.5x^2+xy}{3-2z}\Big|_{x=0}^{x=2}\,dy\,dz \\ &= 10\int_0^1\int_0^2 \frac{2+2y}{3-2z}\,dy\,dz \\ &= 10\int_0^1 \frac{2y+y^2}{3-2z}\Big|_{y=0}^{y=2}\,dz \\ &= 10\int_0^1 \frac{8}{3-2z}\,dz \\ &= 10\left(-4\ln(3-2z)\big|_0^1\right) \\ &= 40\ln(3) \approx 44\text{kg} \end{aligned}$$

There are six basic types of solids (sometimes a solid has to be broken into basic pieces):

(1) E: $a \le x \le b$, $g(x) \le y \le h(x)$, $u(x,y) \le z \le v(x,y)$.

$$\int\int\int_E f(x,y,z)dV = \int_a^b \int_{g(x)}^{h(x)} \int_{u(x,y)}^{v(x,y)} f(x,y,z)dzdydx.$$

(2) E: $a \le x \le b$, $g(x) \le z \le h(x)$, $u(x,z) \le y \le v(x,z)$.

$$\int\int\int_E f(x,y,z)dV = \int_a^b \int_{g(x)}^{h(x)} \int_{u(x,z)}^{v(x,z)} f(x,y,z)dydzdx.$$

(3) E: $a \le y \le b$, $g(y) \le x \le h(y)$, $u(x,y) \le z \le v(x,y)$.

$$\int\int\int_E f(x,y,z)dV = \int_a^b \int_{g(y)}^{h(y)} \int_{u(x,y)}^{v(x,y)} f(x,y,z)dzdxdy.$$

(4) E: $a \le y \le b$, $g(y) \le z \le h(y)$, $u(y,z) \le x \le v(y,z)$.

$$\int\int\int_E f(x,y,z)dV = \int_a^b \int_{g(y)}^{h(y)} \int_{u(y,z)}^{v(y,z)} f(x,y,z)dxdzdy.$$

(5) E: $a \le z \le b$, $g(z) \le x \le h(z)$, $u(x,z) \le y \le v(x,z)$.

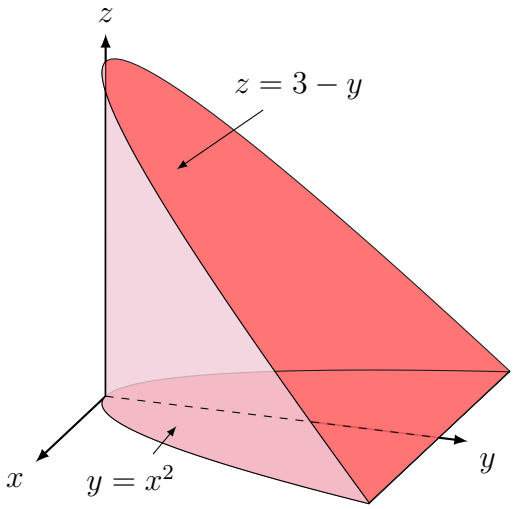
$$\int\int\int_E f(x,y,z)dV = \int_a^b \int_{g(z)}^{h(z)} \int_{u(x,z)}^{v(x,z)} f(x,y,z)dydxdz.$$

(6) E: $a \le z \le b$, $g(z) \le y \le h(z)$, $u(y,z) \le x \le v(y,z)$.

$$\int\int\int_E f(x,y,z)dV = \int_a^b \int_{g(z)}^{h(z)} \int_{u(y,z)}^{v(y,z)} f(x,y,z)dxdydz.$$

If the integrand is 1 then the triple integral finds the volume of E. Also, the average value of $f(x,y,z)$ over E is $\dfrac{1}{V}\iiint_E f(x,y,z)\,dV$ where V is the volume of E.

Let's find the average value of $f(x,y,z) = x^2$ over the solid E inside the parabolic cylinder $y = x^2$ between the planes $z = 3 - y$ and $z = 0$.



First let's calculate the volume (observing the symmetry across the yz-plane).

$$\begin{aligned}
V = \iiint_E 1\,dV &= 2\int_0^{\sqrt{3}}\int_{x^2}^{3}\int_0^{3-y} 1\,dz\,dy\,dx \\
&= 2\int_0^{\sqrt{3}}\int_{x^2}^{3} 3-y\,dy\,dx \\
&= 2\int_0^{\sqrt{3}} 3y - \frac{1}{2}y^2\Big|_{y=x^2}^{y=3}\,dx \\
&= 2\int_0^{\sqrt{3}} \frac{9}{2} - 3x^2 + \frac{1}{2}x^4\,dx \\
&= 2\left(\frac{9}{2}(\sqrt{3}) - (\sqrt{3})^3 + \frac{1}{10}(\sqrt{3})^5\right) = 2\sqrt{3}\left(\frac{9}{2} - 3 + \frac{9}{10}\right) = \frac{24\sqrt{3}}{5}.
\end{aligned}$$

Now let's calculate the triple integral of $f(x,y,z) = x^2$ over the solid region E (observing the symmetry across the yz-plane and that f is an even function with respect to x).

$$\begin{aligned}
V = \iiint_E x^2\,dV &= 2\int_0^{\sqrt{3}}\int_{x^2}^{3}\int_0^{3-y} x^2\,dz\,dy\,dx \\
&= 2\int_0^{\sqrt{3}}\int_{x^2}^{3} x^2(3-y)\,dy\,dx \\
&= 2\int_0^{\sqrt{3}} x^2\left(3y - \frac{1}{2}y^2\Big|_{y=x^2}^{y=3}\right)dx \\
&= 2\int_0^{\sqrt{3}} \frac{9}{2}x^2 - 3x^4 + \frac{1}{2}x^6\,dx \\
&= 2\left(\frac{3}{2}(\sqrt{3})^3 - \frac{3}{5}(\sqrt{3})^7 + \frac{1}{14}(\sqrt{3})^7\right) = 2\sqrt{3}\left(\frac{9}{2} - \frac{27}{5} + \frac{27}{14}\right) = \frac{72\sqrt{3}}{35}.
\end{aligned}$$

So the average value of $f(x,y,z) = x^2$ over the solid region E is $\dfrac{72\sqrt{3}/35}{24\sqrt{3}/5} = \dfrac{3}{7}$.

What if we decided to evaluate the integral above in the $dy\,dx\,dz$ order instead? Well, the bounds would be $0 \le z \le 3$, $-\sqrt{3-z} \le x \le \sqrt{3-z}$ and $x^2 \le y \le 3 - z$.

$$\begin{aligned}
V = \iiint_E x^2\,dV &= 2\int_0^{3}\int_0^{\sqrt{3-z}}\int_{x^2}^{3-z} x^2\,dy\,dx\,dz \\
&= 2\int_0^{3}\int_0^{\sqrt{3-z}} x^2(3-z) - x^4\,dx\,dz \\
&= 2\int_0^{3} \frac{1}{3}(\sqrt{3-z})^3(3-z) - \frac{1}{5}(\sqrt{3-z})^5\,dz \\
&= 2\int_0^{3} \frac{1}{3}(3-z)^{5/2} - \frac{1}{5}(3-z)^{5/2}\,dz \\
&= 2\int_0^{3} \frac{2}{15}(3-z)^{5/2}\,dz \\
&= 2\left(-\frac{4}{105}(3-z)^{7/2}\Big|_0^3\right) = 2\,\frac{4(27\sqrt{3})}{105} = \frac{72\sqrt{3}}{35}.
\end{aligned}$$